

ATAR course Year 12 Physics

Practice Exam 1 (Unit 3 & 4) Answers

Section one: Short answer

(57 marks)

Question 1

(4 marks)

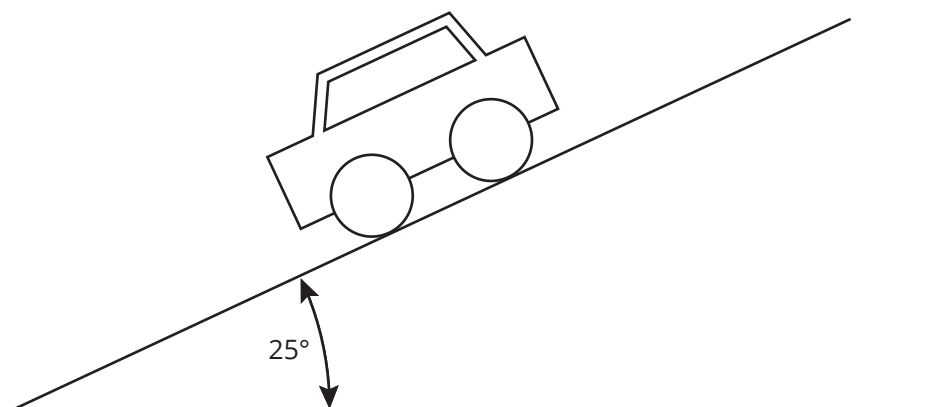
Gerald and Omar are standing on a first-floor balcony. Gerald throws a ball horizontally and at the same instant Omar drops an identical ball. Sally, who is standing below, observes the balls hit the ground at the same time. Explain this observation. Identify the relevant mathematical relationship in your explanation.

Description	Marks
Recognition that the time taken to fall is determined only by the balls' motion in the vertical direction	1
Recognition that both have the same initial and final velocity in the vertical direction (and travel the same vertical distance)	1
Recognition that both experience the same acceleration due to gravity in the vertical direction (i.e. 9.80 m s^{-2})	1
Recognition that application of $v = u + at$ will result in the same time of flight for the two balls (or $s = ut + \frac{1}{2}at^2$)	1
Total	4

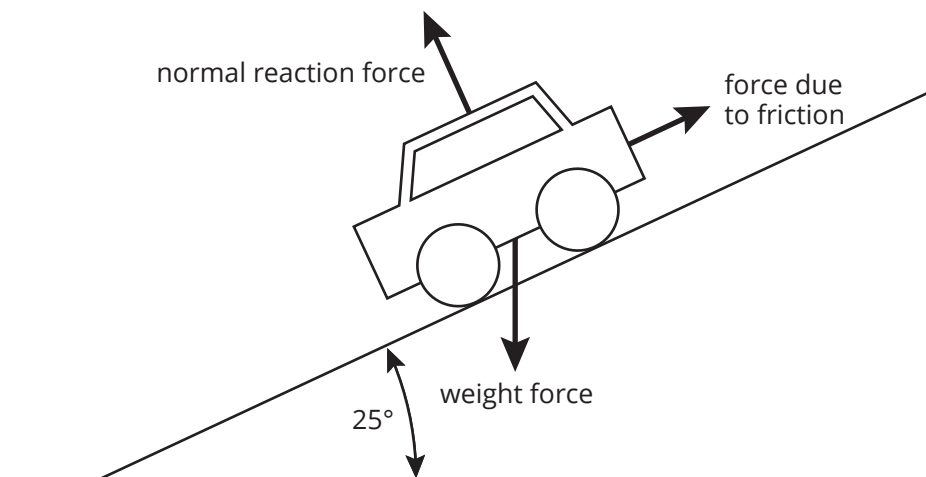
Question 2

(5 marks)

A car of mass 1540 kg is stationary on a hill that has a slope of 25° as represented below.



Sketch and label the forces acting on the stationary car on the diagram given, and calculate the force needed to stop the car from moving.

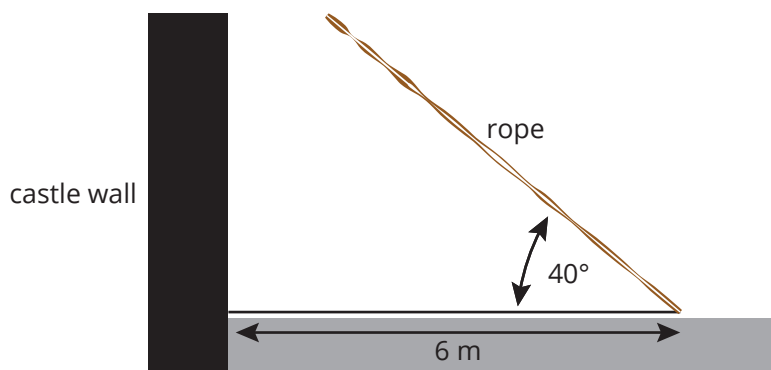


Description	Marks
1 mark for each force	1-3
$F = m g \sin 25^\circ$ $= 1540 \times 9.80 \times \sin 25^\circ$	1
$F = 6.38 \times 10^3 \text{ N}$	1
Total	5

Question 3

(4 marks)

An 8 m castle drawbridge of uniform mass 3200 kg is raised by a rope attached at an angle of 40° and 6 m from its end as shown below. Determine the tension (force) in the rope needed to begin moving the drawbridge from its horizontal position.

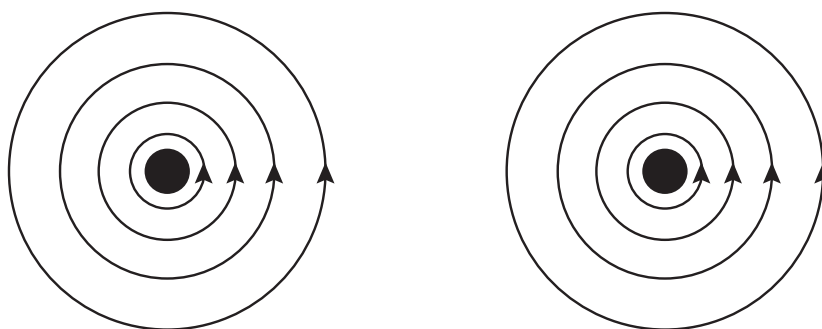


Description	Marks
$\Sigma \text{ anticlockwise moments} = \Sigma \text{ clockwise moments}$	1
$T_{\text{rope}} \times 6 \times \sin 40^\circ = 3200 \times 9.8 \times 4.0$	1
$T_{\text{rope}} = \frac{1.2544 \times 10^5}{3.857}$ $= 3.25 \times 10^4 \text{ N}$	1-2
Total	4

Question 4

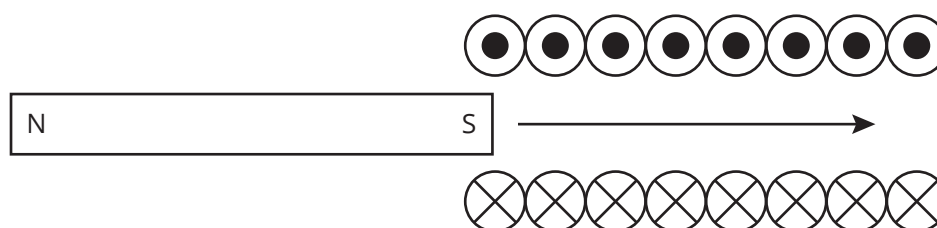
(5 marks)

- a Complete the diagram below by drawing at least **five** field lines around the conductors and state whether the wires are attracted towards or repelled from each other. (3 marks)



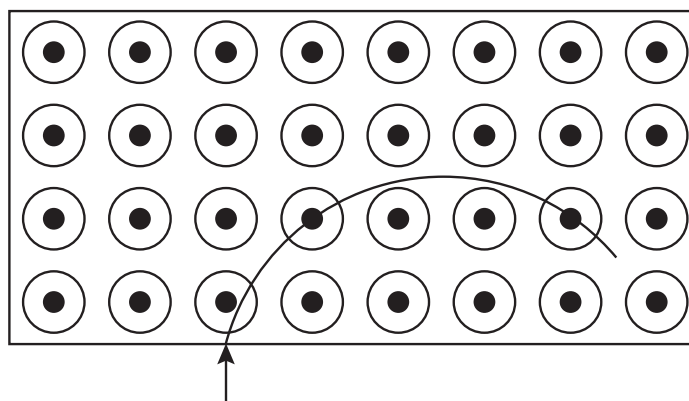
Description	Marks
Anticlockwise field lines around each wire	1
Field strength represented by lines showing greater separation from each other as they move away from the conductors	1
Force of attraction between wires	1
Total	3

- b Complete the diagram below by showing the direction of movement of the magnet that generates the current in the solenoid represented below. (1 mark)



Description	Marks
Magnet moving into solenoid (i.e. to the right)	1
Total	1

- c Complete the diagram by drawing the path a positively charged particle will take on entering the magnetic field shown below. (1 mark)



Description	Marks
Path of particle shown going to the right	1
Total	1

Question 5**(4 marks)**

What is the wavelength of the radiation emitted from 50 Hz electric power lines? What is the energy of a photon of this wavelength?

Description	Marks
$c = f\lambda$ $\lambda = \frac{c}{f}$	1
$\lambda = \frac{3.00 \times 10^8}{50} = 6.00 \times 10^6 \text{ m}$	1
$E = hf$ $E = 6.63 \times 10^{-34} \times 50 = 3.32 \times 10^{-32}$	1-2
Total	4

Question 6**(4 marks)**

Einstein's special theory of relativity is based on the two principles that:

- the speed of light in a vacuum is an absolute constant, and
- all inertial frames of reference are the same.

Identify the apparent contradiction of accepting these two principles and explain how Einstein's special theory of relativity resolves this apparent contradiction.

Description	Marks
Recognition that the equivalence of all inertial frames of reference means observers in different frames of reference moving relative to each other will measure different velocities for the same object, which suggests the speed of light should not be the same when observed from frames of reference moving relative to each other	1-2
Recognition that this apparent contradiction is resolved by:	
• time dilating in the moving frame of reference (i.e. time appears to move slower in a moving frame of reference), and	1
• a moving object will appear shorter, or appear to travel less distance in the moving frame of reference	1
Total	4

Question 7**(5 marks)**

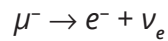
The table below gives leptons and their antiparticles.

Lepton	Lepton symbol	Antilepton symbol
electron	e^-	e^+
electron neutrino	ν_e	$\bar{\nu}_e$
muon	μ^-	μ^+
muon neutrino	ν_μ	$\bar{\nu}_\mu$
tau	τ^-	τ^+
tau neutrino	ν_τ	$\bar{\nu}_\tau$

All leptons have a lepton number of positive one (i.e. $L = +1$) and antileptons have a lepton number of negative one (i.e. $L = -1$).

Use the above information to answer the following questions.

- a Determine whether the following decay for a muon is possible. Explain your answer. (3 marks)



Description	Marks
This decay is not possible.	1
Before decay L = +1	1
After decay L = (+1) + (+1) = +2	1
Total	3

- b When a muon decays to **three** other leptons and/or antileptons, what is the only possible combination? Give your reasoning. (2 marks)

Note: You are not required to identify a specific set of leptons and/or antileptons as decay products but rather a general statement about the number of leptons and/or antileptons permissible when a muon decays to a total of three products.

Description	Marks
The lepton number before and after decay must always be +1.	1
The only possible combination of three decay products must therefore be two leptons and one antilepton, to give (+1) + (+1) + (-1) = +1.	1
Total	2

Question 8

(5 marks)

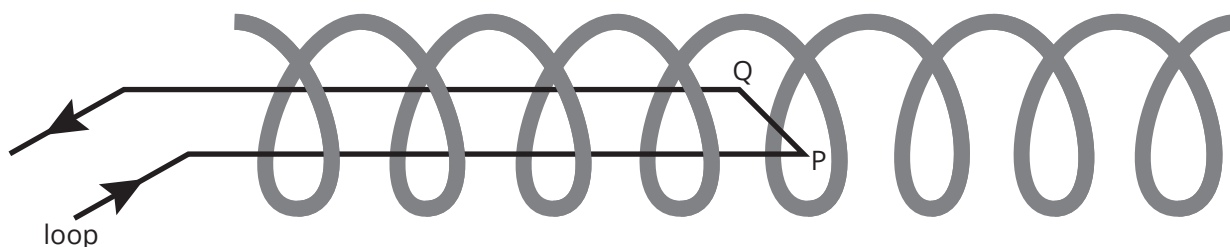
The electric trains in Perth use some of their electric motors as generators when breaking. Briefly describe how an electric generator produces a current and explain why turning the current off to the motor and allowing it to operate as a generator helps slow the train.

Description	Marks
A coil and magnetic field are rotating relative to each other.	1
This creates a changing flux in the coil and induces an EMF (or current).	1
The current generated creates a secondary magnetic field.	1
This secondary magnetic field opposes the external magnetic field (Lenz's law).	1
The interaction between the two magnetic fields opposes the rotation of the wheels.	1
Total	5

Question 9

(4 marks)

A conductor loop is positioned inside a solenoid with its long axis parallel to the axis of the solenoid, as shown in the diagram below. The short side PQ is at right angles to the axis and is 25 mm long. The current in the loop is 5.0 A, while the current in the solenoid produces a magnetic field inside the solenoid with flux density of 40.0 mT to the right.



- a What is the direction of the magnetic force on the side PQ? (1 mark)

Description	Marks
Downwards	1
Total	1

- b What is the magnitude of this force on the side P? (3 marks)

Description	Marks
$F = I\ell B = 5 \times 0.025 \times 0.04$	1-2
$= 5.00 \times 10^{-3} \text{ N}$	1
Total	3

Question 10

(5 marks)

The Australian synchrotron is a light source synchrotron. It accelerates electrons to high speed and then converts this energy to a range of different wavelength electromagnetic radiation (light) to conduct experiments.

- a The synchrotron light is produced when the electrons are forced to change direction by an external magnetic field. Explain how a magnetic field causes the electrons to change direction. (2 marks)

Description	Marks
Moving electrons create their own magnetic field (moving charge induces a magnetic field).	1
The external magnetic field will interact with the field induced by the moving electron to change its direction.	1
Total	2

- b If an electron accelerated to have 2.5 GeV of energy has 0.01% of its energy converted to light (electromagnetic radiation), what is the wavelength of the light? (3 marks)

Description	Marks
100% energy in J = $2.5 \times 10^9 \times 1.60 \times 10^{-19} = 4.0 \times 10^{-10} \text{ J}$	1
0.01% energy in J = $4.0 \times 10^{-14} \text{ J}$	1
$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{4.0 \times 10^{-14}} = 4.97 \times 10^{-12} \text{ m}$	1
Total	3

Question 11**(4 marks)**

What is the wavelength of an electron accelerated to 0.85 the speed of light (i.e. $0.85c$)?

Description	Marks
$p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$	1
$p = \frac{9.11 \times 10^{-31} \times 0.85 \times 3.00 \times 10^8}{\sqrt{1 - \frac{(0.85c)^2}{c^2}}} = \frac{2.323 \times 10^{-22}}{\sqrt{0.2775}} = 4.41 \times 10^{-22} \text{ kg m s}^{-1}$	1-2
$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{4.41 \times 10^{-22}} = 1.5 \times 10^{-12} \text{ m}$	1
Total	4

Question 12**(4 marks)**

The energy level diagram for hydrogen shown below can be used to answer the following question.

E_6	-----	13.60 eV
E_5	=====	13.06 eV
E_4	=====	12.75 eV
E_3	=====	12.09 eV
E_2	=====	10.20 eV
E_1	=====	0.00 eV

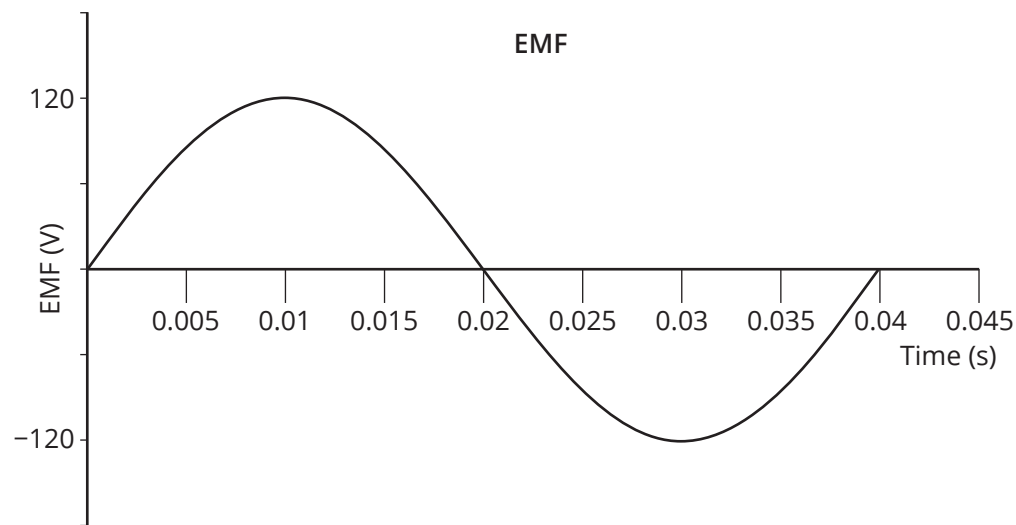
A satellite equipped with an instrument to measure radiation emitted in the ultraviolet region of the electromagnetic spectrum records a wavelength of $9.76 \times 10^{-8} \text{ m}$. Between which two energy levels must the hydrogen atom's electron transition to produce this line?

Description	Marks
$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{9.76 \times 10^{-8}} = 2.037 \times 10^{-18} \text{ J}$	1-2
Energy in eV = $\frac{2.037 \times 10^{-18}}{1.60 \times 10^{-19}} = 12.74 \text{ eV}$	1
Transition will be between energy levels 4 and 1.	1
Total	4

Question 13

(4 marks)

On the axes below, draw a graph to show the EMF for an AC generator with a frequency of 25 Hz and root mean square voltage of 85 V.

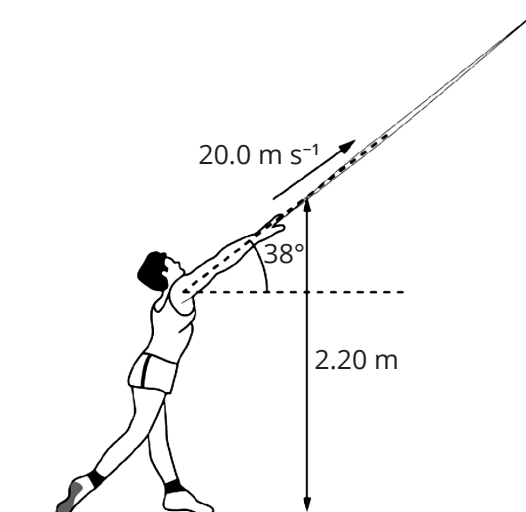


Description	Marks
$T = \frac{1}{25} = 0.04 \text{ s}$	1
$V_{\text{max}} = V_{\text{rms}} \sqrt{2} = 85\sqrt{2} = 120 \text{ V}$	1
Sinusoidal curve	1
Axes labelled	1
Total	4

Question 14

(13 marks)

During a school athletics competition, an athlete threw a javelin as shown below.



At the moment it left the athlete's hand, the 0.85 kg javelin was at a height of 2.2 m above the ground and travelling at 20 m s⁻¹ at an angle of elevation of 38°.

- a** What was the maximum height above the ground reached by the javelin? Ignore the effects of air resistance. (3 marks)

Description	Marks
For the vertical component of the javelin's motion $u_v = 20 \times \sin 38^\circ = 12.31 \text{ m s}^{-1}$ $v_v = 0 \text{ m s}^{-1}$ $a = -9.80 \text{ m s}^{-2}$	1
$v_v^2 = u_v^2 + 2as$ $0 = 12.31^2 + 2 \times (-9.80)s$ $s = \frac{12.31^2}{2 \times 9.80} = 7.74 \text{ m}$	1
Max height = 2.20 + 7.74 = 9.94 m	1
Total	3

- b** How far horizontally from its point of release did the javelin land? Ignore the effects of air resistance. (5 marks)

Description	Marks
Time to reach maximum height: $u_v = 12.31 \text{ m s}^{-1}$ $v_v = 0 \text{ m s}^{-1}$ $a = -9.80 \text{ m s}^{-2}$	1
$v = u + at$ $0 = 12.31 + (-9.80)t$ $t = \frac{12.31}{9.80} = 1.26 \text{ s}$	1

Time to travel from max height to ground: $u_v = 0 \text{ m s}^{-1}$ $s = 9.94 \text{ m}$ $a = 9.80 \text{ m s}^{-2}$	1
$s = ut + \frac{1}{2} at^2$ $9.94 = 0 + 4.90t^2$ $t = \sqrt{\frac{9.94}{4.9}} = 1.42 \text{ s}$	1
For the horizontal component of the javelin's motion and hence its range: $t = 1.26 + 1.42 = 2.68 \text{ s}$ $v_h = 20 \cos 38^\circ = 15.76 \text{ m s}^{-1}$ $s = v_h \times t = 15.76 \times 2.68 = 42.2 \text{ m}$	1
Total	5

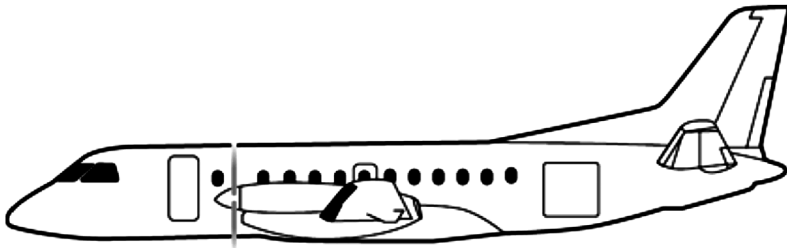
- c** How much kinetic energy does the javelin have when it strikes the ground at the end of its flight? (5 marks)

Description	Marks
Recognition that when the javelin leaves the athlete's hand, it has both kinetic and potential energy: KE due to its motion and PE due to being 2.2 m above ground	1
Recognition that when the javelin strikes the ground, all its energy will be kinetic energy and: KE + PE when javelin leaves athlete's hand = KE when it strikes ground	1
KE when leaving athlete's hand: $E_k = \frac{1}{2} mv^2$ $= \frac{1}{2} \times 0.85 \times (20)^2$ $= 170 \text{ J}$	1
PE when leaving athlete's hand: $E_p = mg\Delta h = 0.85 \times 9.80 \times 2.20 = 18.3 \text{ J}$	1
Thus, total KE when it strikes the ground = $170 + 18.3 = 188 \text{ J}$	1
Total	5

Question 15

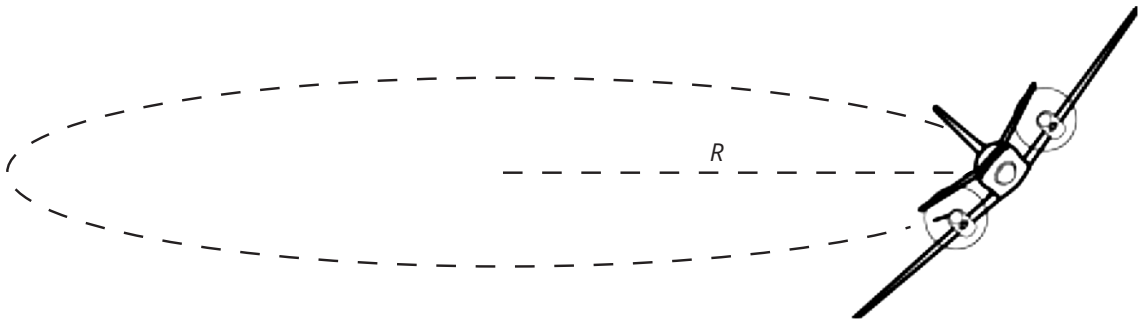
(14 marks)

A pilot of mass 75 kg was flying in an aeroplane as shown below.



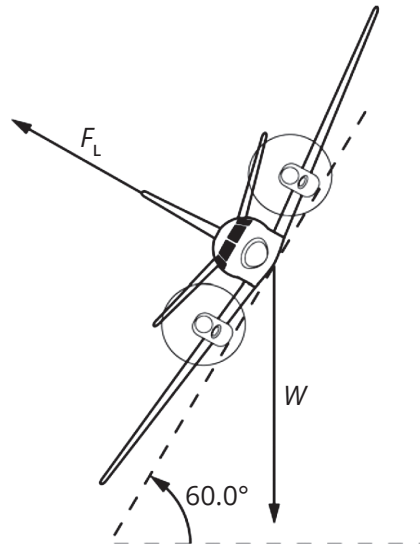
The combined mass of the pilot and aeroplane was 5000 kg.
Initially, the pilot flew in a horizontal circle as shown by the diagrams in parts a and b of this question.

- a While flying in a horizontal circle, as represented below, the aeroplane flew at a *constant* speed of 120 m s⁻¹ and it took 20 seconds for it to complete one full circle. What was the radius, *R*, of the horizontal circle that the aeroplane flew around? (2 marks)



Description	Marks
$v = \frac{2\pi r}{T}$ Thus, $r = \frac{vT}{2\pi}$	1
$r = \frac{120 \times 20}{2\pi} = 382 \text{ m}$	1
Total	2

- b** The figure below shows the only two unbalanced forces that were acting on the aeroplane during the manoeuvre to fly the horizontal circle shown in part a of this question. Calculate the magnitude of the lift force, F_L . (3 marks)

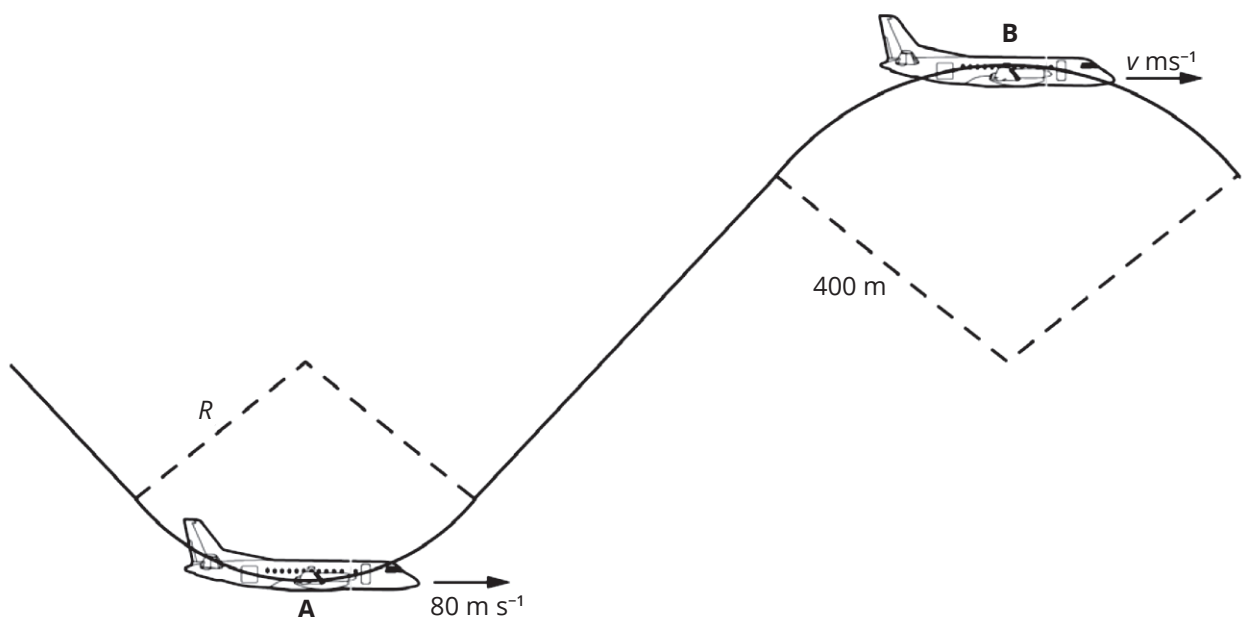


Description	Marks
$W = mg = 5000 \times 9.80 = 4.90 \times 10^4 \text{ N}$	1
$\cos 60^\circ = \frac{W}{F_L}$ Thus, $F_L = \frac{W}{\cos 60^\circ}$	1
$F_L = \frac{4.9 \times 10^4}{\cos 60^\circ} = 9.8 \times 10^4 \text{ N}$	1
Total	3

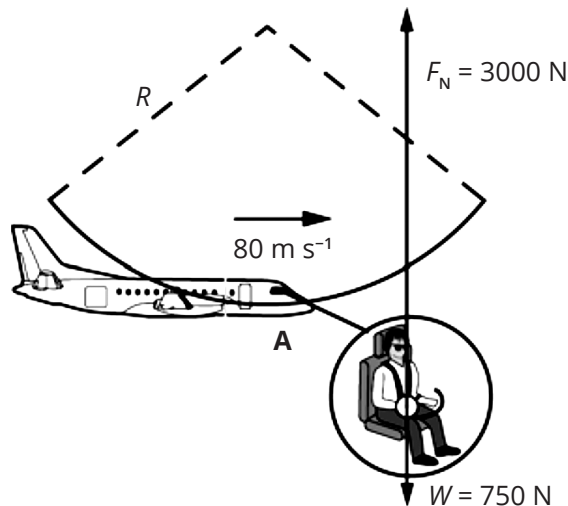
A short time later, the pilot performed the aerobatics as represented in the figure below. This time the manoeuvres involved flying along the arcs of two vertical circles shown at A and B.

A shows the aeroplane at the bottom of an arc of radius R flying at a constant horizontal speed of 80 m s^{-1} .

B shows the aeroplane at the top of an arc of radius 400 metres flying at an unknown constant horizontal speed of $v \text{ m s}^{-1}$.



- c At the point A as shown below, the pilot's apparent weight is 3000 N. Draw and label the direction and magnitude of the forces acting on the pilot at this instant on the diagram provided. (2 marks)



Description	Marks
$W = 750 \text{ N}$	1
$F_N = 3000 \text{ N}$	1
Total	2

- d If the aeroplane was flying at a constant speed of 80 m s^{-1} at point A, what would be the radius, R , of the arc flown? (4 marks)

Description	Marks
The centripetal force: $F_c = F_N - W = 3000 - (75 \times 9.80) = 2.4075 \times 10^3 \text{ N}$	1-2
$F_c = \frac{mv^2}{r}$ Thus, $r = \frac{mv^2}{F_c}$	1
$r = \frac{75 \times (80)^2}{2.4075 \times 10^3} = 199 \text{ m}$	1
Total	4

- e At point B the pilot experiences 'apparent weightlessness' or freefall. At what speed would the aeroplane need to be flying at point B in order for the pilot to experience 'apparent weightlessness'? (3 marks)

Description	Marks
Recognition that for apparent weightlessness the pilot will experience only acceleration due to gravity	1
$a = \frac{v^2}{r}$ Thus, $v^2 = ar$	1
$v^2 = 9.8 \times 400 = 3920$ $v = \sqrt{3920} = 62.6 \text{ m s}^{-1}$	1
Total	3

Question 16

(12 marks)

A weather satellite is placed in orbit 600 km above the Earth's surface.

- a** What orbital velocity must this satellite have? (3 marks)

Description	Marks
$\frac{m_{\text{sat}} v^2}{r} = \frac{G m_{\text{sat}} m_E}{r^2}$ $\text{Thus, } v = \sqrt{\frac{G m_E}{r}}$	1
$r = 6.38 \times 10^6 + 6 \times 10^5 = 6.98 \times 10^6 \text{ m}$	1
$v = \sqrt{\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.98 \times 10^6}}$ $= 7.55 \times 10^3 \text{ m s}^{-1}$	1
Total	3

- b** Calculate the period of orbit of the satellite. (2 marks)

Description	Marks
$v = \frac{2\pi r}{T}$ $\text{Thus, } T = \frac{2\pi r}{v}$	1
$T = \frac{2\pi \times 6.98 \times 10^6}{7550} = 5.81 \times 10^3 \text{ s}$	1
Total	2

- c** What acceleration due to the Earth's gravitational field does the satellite experience? (3 marks)

Description	Marks
$a = \frac{v^2}{r}$	1
$a = \frac{(7550)^2}{6.98 \times 10^6} = 8.17 \text{ m s}^{-2}$	1-2
Total	3

- d** Why do objects in the satellite experience weightlessness? (2 marks)

Description	Marks
Objects in the satellite accelerate towards the Earth at the same rate as the satellite.	1
Thus, as weight is a reaction force, objects in the satellite experience no reaction force.	1
Total	2

- e Orbiting satellites are fitted with small propulsion devices to assist in stabilising their orbit. Without these devices, they are likely to slow. Explain why the rockets may slow and what would happen if they did. (2 marks)

Description	Marks
Celestial bodies other than the Earth can have a gravitational influence on the satellite, which can slow it (also, the Earth's gravitational field is not uniform so its influence varies as the satellite circles the Earth).	1
If the satellite slows, it will eventually spiral into the Earth.	1
Total	2

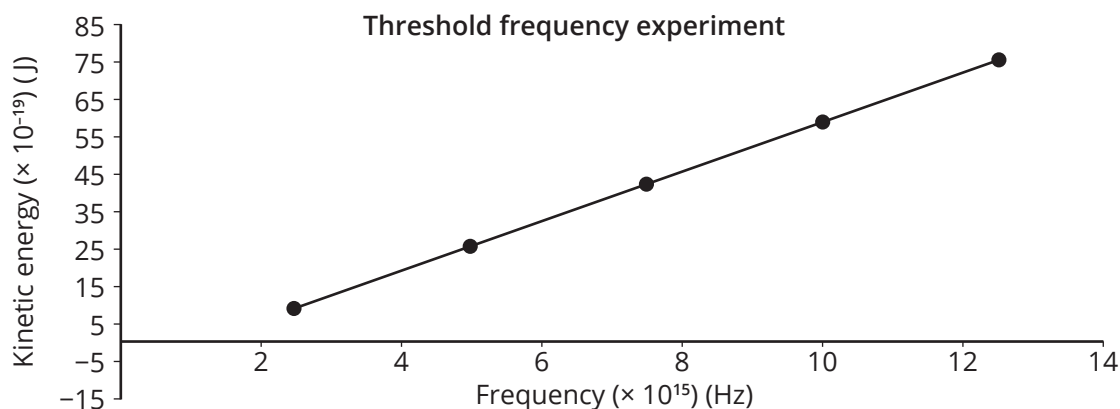
Question 17

(14 marks)

In an experiment to determine the threshold frequency and work function for a particular metal, a researcher measured the kinetic energy of the electrons ejected from the metal's surface at a range of frequencies. The results are tabulated below.

Frequency ($\times 10^{15}$) (Hz)	Kinetic energy of electron ($\times 10^{-19}$) (J)
2.50	9.055
5.00	25.63
7.50	42.20
1.00	58.78
1.25	75.36

- a Plot a graph of frequency against kinetic energy on the axis provided below. (5 marks)



Description	Marks
Frequency on x-axis	1
Axes labelled with units	1
Appropriate scales on axes	1
Accurate plot	1
Points joined with a line	1
Total	5

- b** Use data from the graph to determine the threshold frequency for the metal. (2 marks)

Description	Marks
Recognition that threshold frequency is found by extending the graph back to intercept with x-axis	1
Value is 1.1×10^{15} Hz (accept values $1-1.3 \times 10^{15}$ Hz)	1
Total	2

- c** Use data from the graph to determine the work function for the metal. (2 marks)

Description	Marks
Recognition that work function is found by extending the graph back to intercept with y-axis	1
Value is 7.52×10^{-19} J (accept values $7-8 \times 10^{-19}$ J)	1
Total	2

- d** Determine the slope of the line. (2 marks)

Description	Marks
Uses the relationship $\frac{\Delta y}{\Delta x}$ to find slope	1
Value should be 6.63×10^{-34}	1
Total	2

The researcher carries out the same experiment to determine the threshold frequency and work function for a second metal, and graphs the frequency against kinetic energy.

- e** Predict the slope of the line for this second metal. (1 mark)

Description	Marks
Slope will be 6.63×10^{-34}	1
Total	1

- f** Explain what the effect studied in this experiment provides evidence for in respect of electromagnetic radiation. (2 marks)

Description	Marks
This phenomenon is evidence for the particle nature of light (EMR).	1
It suggests light is quantised into discrete packets of energy.	1
Total	2

Question 18

(13 marks)

A space probe is sent to make observations of a planet orbiting a star in a neighbouring galaxy. The planet is 24.6 light-years from Earth and the probe is sent from Earth with a speed $0.82c$. The probe carries an atomic clock that was synchronised with a similar clock on Earth at the start of the journey.

- a** How long will the journey take as measured by the clock on Earth? (2 marks)

Description	Marks
Time on Earth clock: $t_{ec} = \frac{s}{v} = \frac{24.06c}{0.82c}$	1
$t_{ec} = 30.0$ years	1
Total	2

- b** How long will the journey take as measured by the clock on the probe? (2 marks)

Description	Marks
$t_{pc} = t_{ec} \sqrt{1 - \frac{v^2}{c^2}} = 30\sqrt{1 - (0.82^2)}$	1
$t_{pc} = 17.2$ years	1
Total	2

- c** From the space probe's frame of reference, what is the distance travelled to reach the planet? (2 marks)

Description	Marks
$\ell = \ell_{ep} \sqrt{1 - \frac{v^2}{c^2}} = 24.6\sqrt{1 - (0.82^2)}$	1
$\ell = 14.1$ light-years	1
Total	2

- d** From the space probe's frame of reference, at what velocity is it travelling? (2 marks)

Description	Marks
$v = \frac{s}{t} = \frac{14.1c}{17.2}$	1
$v = 0.82c$	1
Total	2

- e** On arrival, the probe launches a smaller probe towards the star around which the planet orbits at a speed of $0.650c$. What is the speed of this smaller probe from the Earth's frame of reference? (2 marks)

Description	Marks
$u = \frac{v + u'}{1 + \frac{vu'}{c^2}} = \frac{0.82c + 0.65c}{1 + \frac{0.82c \times 0.65c}{c^2}}$	1
$u = 0.959c$	1
Total	2

- f** On arrival at the planet, the space probe sends signals of its measurements back to Earth using various parts of the electromagnetic spectrum.
- i** At what velocity will the signals travel from the probe's frame of reference? (1 mark)

Description	Marks
c ($3.00 \times 10^8 \text{ m s}^{-1}$)	1
Total	1

- ii** At what velocity will the signals travel from the Earth's frame of reference? (1 mark)

Description	Marks
c ($3.00 \times 10^8 \text{ m s}^{-1}$)	1
Total	1

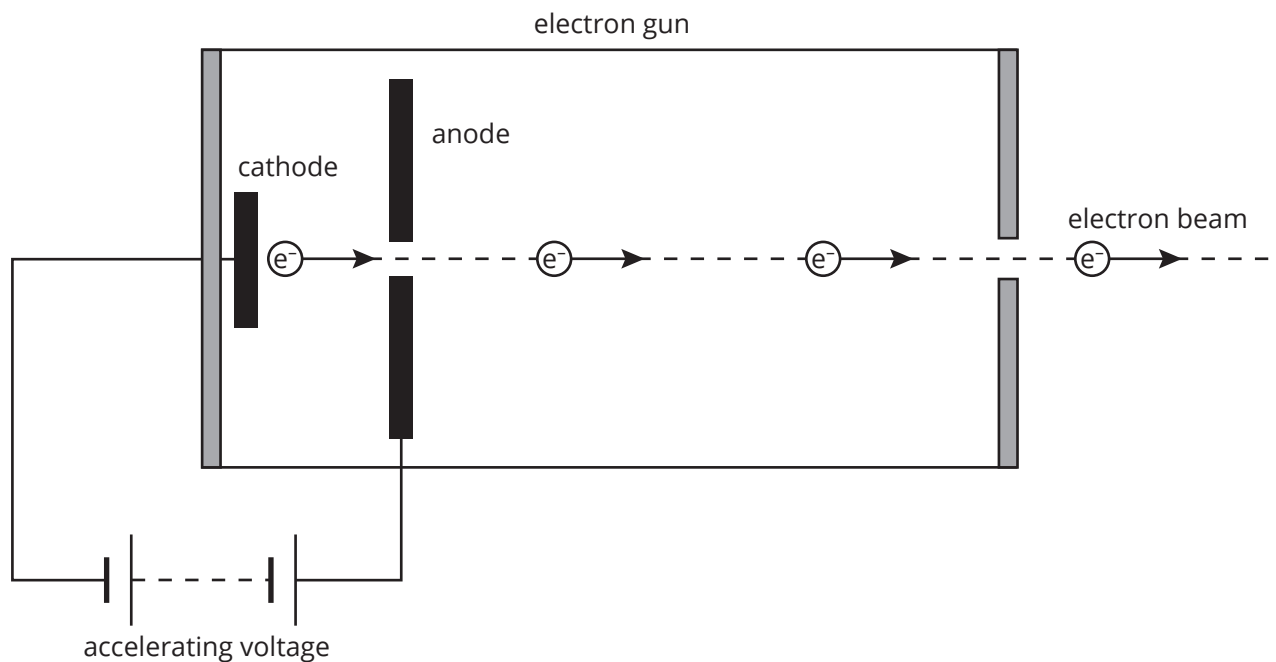
- iii** Explain your answers to (i) and (ii). (1 mark)

Description	Marks
The speed of EMR is a constant at $3.00 \times 10^8 \text{ m s}^{-1}$ in all frames of reference.	1
Total	1

Question 19

(16 marks)

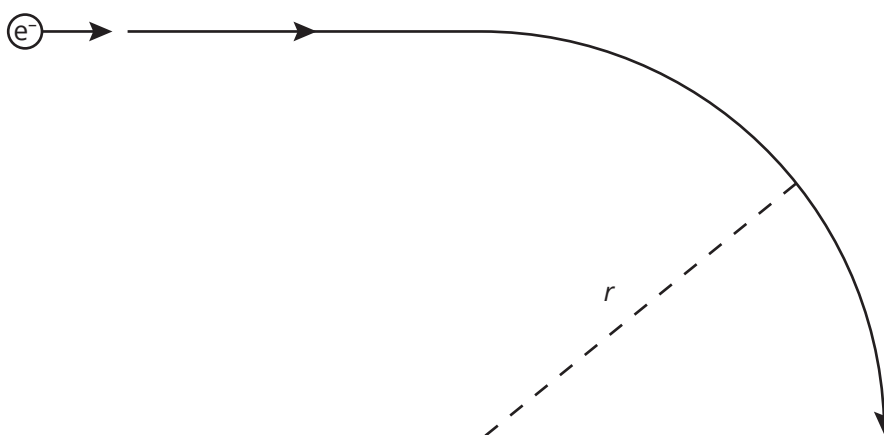
An electron gun is used to inject electrons into a synchrotron. The electron gun releases electrons from its cathode, which are then accelerated across a potential difference of 16 kV. The distance between the cathode and anode is 25 cm. Below is shown a schematic diagram of an electron gun.



- a With what speed will the electrons exit the electron gun? (4 marks)

Description	Marks
$E = W = Vq = 16\,000 \times 1.60 \times 10^{-19}$	1
$E = 2.56 \times 10^{-15} \text{ J}$	1
$E = \frac{1}{2}mv^2$ $v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{2 \times 2.56 \times 10^{-15}}{9.11 \times 10^{-31}}}$	1
$v = 7.50 \times 10^7 \text{ m s}^{-1}$	1
Total	4

The electron gun's accelerating voltage was adjusted so that the exiting electrons were travelling at a speed of $1.5 \times 10^5 \text{ m s}^{-1}$. The path taken by the electrons as they entered a magnetic field of strength 20 mT is shown below.



- b State the direction of the magnetic field that causes the electrons to follow the path shown above. (1 mark)

Description	Marks
Into the page	1
Total	1

- c What is the magnitude of the force acting on the electrons as they travel through the magnetic field? (2 marks)

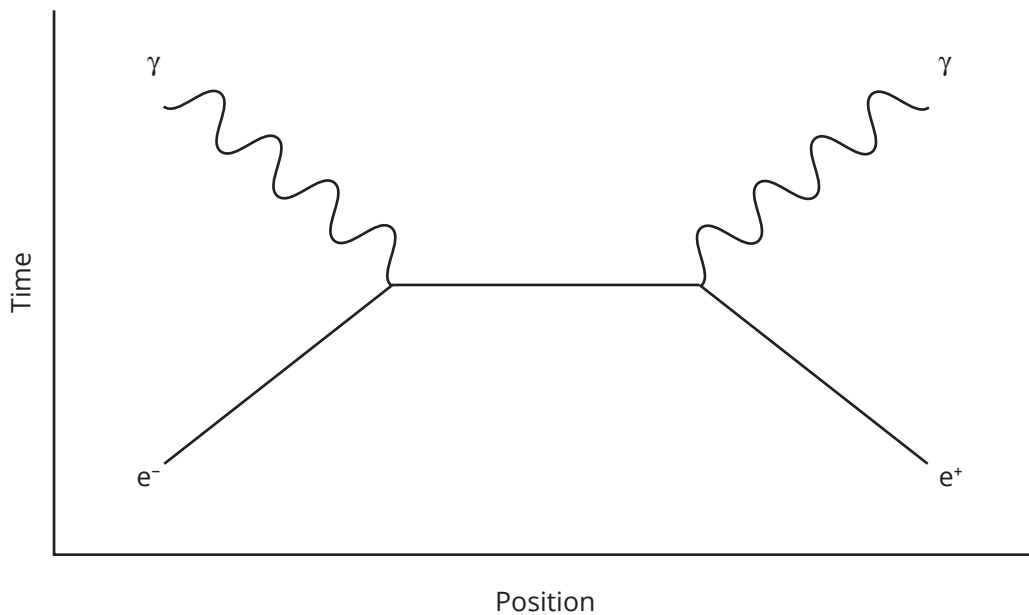
Description	Marks
$F = qvB = 1.60 \times 10^{-19} \times 1.5 \times 10^5 \times 20 \times 10^{-3}$	1
$F = 4.80 \times 10^{-16} \text{ N}$	1
Total	2

- d What is the radius of the arc of the circle that the electrons follow while they are in the magnetic field? (2 marks)

Description	Marks
$r = \frac{mv}{qB} = \frac{9.11 \times 10^{-31} \times 1.5 \times 10^5}{1.60 \times 10^{-19} \times 20 \times 10^{-3}}$	1
$r = 4.27 \times 10^{-5} \text{ m}$	1
Total	2

An experiment is carried out in the synchrotron where an electron collides with a positron (the antiparticle of an electron).

- e The product of a low-energy collision is two photons. Draw a Feynman diagram to represent this collision. Label the diagram. (3 marks)



Description	Marks
Electron and positron shown with direction	1
Photons represented with waves	1
Time and position axes labelled	1
Total	3

- f To produce particles heavier than themselves, an electron and a positron need to collide at high kinetic energies. When they collide at low energy, they produce only two (or more) photons. Explain why high energies are needed for production of particles heavier than themselves. (2 marks)

Description	Marks
Particles emerging from a collision have mass and velocity such that there is conservation of momentum.	1
The higher velocity of colliding particles increases the chance of a high-mass particle emerging from the collision.	1
Total	2

- g Another consequence of high-energy collisions is greater resolution to study finer structures. Explain how high-energy collisions provide higher resolution. (2 marks)

Description	Marks
High-energy collisions have high momentum.	1
So the wavelength of the particle is less, as shown by de Broglie's wavelength formula: $\lambda = \frac{h}{mv} = \frac{h}{p}$	1
Total	2

Section three: Comprehension

(37 marks)

Question 20

(20 marks)

'X' for unknown

While working on cathode rays (high-energy electron beams) and their effects in 1895, Wilhelm Roentgen observed that when electrons accelerated to a high speed in a vacuum tube struck the wall of the glass tube, fluorescent minerals some distance away would glow. It was known electrons travelled only a few centimetres in air so Roentgen concluded that these effects were due to a new type of radiation he called 'X-radiation' or X-rays. Because these X-rays were not detectable by electric or magnetic fields, it was concluded they were not charged particles.

About a month after Roentgen's discovery, J. J. Thompson observed that if the X-rays were passed through a gas, the gas became electrically conducting.

It was suggested the X-rays were a form of invisible light but no refraction, interference or diffraction patterns were observed when the rays were used with ordinary optical instruments. It was noted that the available diffraction gratings at that time had a typical width of the order of 10^{-6} m. In approximately 1912, it was suggested that to test if X-rays were a wave they could be directed onto crystals, as it was estimated that atoms in a crystal were separated by about 10^{-10} m so the crystal could act as a diffraction grating. Scattering of X-rays from crystals did show diffraction patterns suggesting the wave-like nature of X-rays.

It was also observed that X-rays are produced when high-speed electrons strike a metal surface. But it was also shown that electrons are ejected from metals that are exposed to X-rays. The specific frequency of X-rays that cause the ejection of an electron is unique to each metal and more electrons are ejected with higher intensity X-rays of the required frequency.

- a** Explain why mineral exposed to this new radiation would fluoresce. (3 marks)

Description	Marks
The radiation's energy was absorbed by electrons in atoms of the mineral.	1
The electrons would be excited to a higher energy level.	1
The excited electrons would return to lower energy levels, releasing the energy they had absorbed in the visible region of the EMR.	1
Total	3

- b** What **two** conclusions, other than the one given in the information, can be made about X-radiation based on the nature of the interaction of the X-radiation with electric and magnetic fields? (2 marks)

Description	Marks
The radiation may be an uncharged particle, such as a neutron.	1
The radiation may be wave-like.	1
Total	2

- c** Why can it be concluded that the X-radiation is not a charged particle? Explain the physics principle on which this conclusion is based. (2 marks)

Description	Marks
Electric and magnetic fields had no effect on the X-radiation.	1
Moving charged particles create a magnetic field, so they will experience a force from an external magnetic field.	1
Total	2

- d** Explain J. J. Thompson's observation. (3 marks)

Description	Marks
Recognition that the conductivity of the gas was due to the presence of charged particles	1
The radiation's energy was absorbed by electrons in atoms of the gas.	1
These electrons were ejected from the atoms to form ions (or charged particles) (i.e. the atoms were ionised).	1
Total	3

- e** Why were no interference or diffraction patterns observed with the typical gratings available at the time of the discovery of the X-radiation? Estimate the wavelength of X-radiation based on the information provided. Identify the evidence in the supplied information that supports your estimate. (3 marks)

Description	Marks
Interference and diffraction only occur when the slits (or objects in the path of the wave) are of similar size to the wavelength of the waves.	1
The wavelength of X-radiation is about 10^{-10} m.	1
The separation of atoms in crystals is about 10^{-10} m.	1
Total	3

- f** What can be concluded about the nature of X-rays from the observation of the effect of exposing metals to X-rays? What name is given to this effect? (2 marks)

Description	Marks
X-rays are quantised (i.e. have particle-like properties).	1
Ejection of electrons from metals that have X-rays shone on them is an example of the photoelectric effect.	1
Total	2

- g** Explain why high-energy X-rays have a smaller wavelength. Support your answer with the appropriate mathematical relationship. (2 marks)

Description	Marks
There is an inverse relationship between energy of a photon and its wavelength.	1
$\lambda = \frac{hc}{E}$	1
Total	2

- h** Tungsten is often used in X-ray tubes for making X-rays for medical purposes. What is the minimum energy an electron striking a tungsten plate needs to generate an X-ray with a wavelength of 0.0184 nm? (3 marks)

Description	Marks
$\lambda = 0.0184 \text{ nm} = 1.84 \times 10^{-11} \text{ m}$	1
$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.84 \times 10^{-11}}$	1
$= 1.08 \times 10^{-14} \text{ J}$	1
Total	3

Question 21**(17 marks)**

What mass is and how we measure it: the electronic kilogram

Many physicists are grappling with the question of 'What is mass?' In Newtonian physics, mass is the amount of matter in an object and the concept of mass was established by Newton with his recognition of the following relationship.

$$F = ma$$

With the development of quantum physics and relativity, the Newtonian concept of mass was challenged. Einstein's now famous equation relates the rest mass of a particle to its total energy. As our understanding of the quantisation of matter and relativity developed, it was realised that the mass in Einstein's equation might be considered to have a single quantum of energy.

The issue of how we define mass is coupled with a debate about how we measure it. Currently we measure mass by comparing an object to the international prototype kilogram. The standard unit of mass, the kilogram, is defined as being equal to the mass of the international prototype kilogram. This prototype is a platinum-iridium alloy but it has been found that its mass varies due to a number of reasons, including collecting dirt from the atmosphere. Other shortcomings with using an object as a standard are that copies need to be made for use around the world and subunits in smaller sizes (such as 500 g, 100 g and 50 g objects) need to be made. This copying and dividing process introduces errors.

The earliest basis for measuring mass was essentially placing an object of known mass in a pan on one side of a balance arm pivoted at its centre and the object to be weighed in a second pan on the opposite side of the pivot arm of the balance. Modern electronic balances are essentially based on this principle.

To overcome these shortcomings, the International Committee on Weights and Measures is considering a plan to redefine the kilogram on the basis of unchanging quantum properties of nature rather than a physical object. The watt balance is an instrument that may allow this to be done.

The watt balance can express a value for mass in terms of a current measured against the force of gravity. It operates on the principle that a conducting wire of known length carrying a current will experience a force when in a perpendicular magnetic field. The current is varied so that the force experienced by the conductor precisely counteracts the weight force of the mass.

Another type of watt balance, the Kibble balance, does away with the need to measure the length of the conductor and the strength of the magnetic field by moving the conducting wire through the magnetic field at a constant known velocity. By Faraday's law of electromagnetic induction, an EMF is induced across the ends of the conductor. Thus, with accurate measurement of the induced EMF, current, velocity of the conductor and the local gravitational field, an accurate mass can be determined.

The advantage of using a watt balance is that current and voltage are defined in terms of universal physical constants such as the speed of light and Planck's constant; thus, this will provide a definition of the kilogram in terms of these constants rather than a physical object.

- a** Why are physicists trying to find a replacement for the international prototype kilogram? (2 marks)

Description	Marks
The mass of the international prototype kilogram varies.	1
The copying and subdividing process introduces errors.	1
Total	2

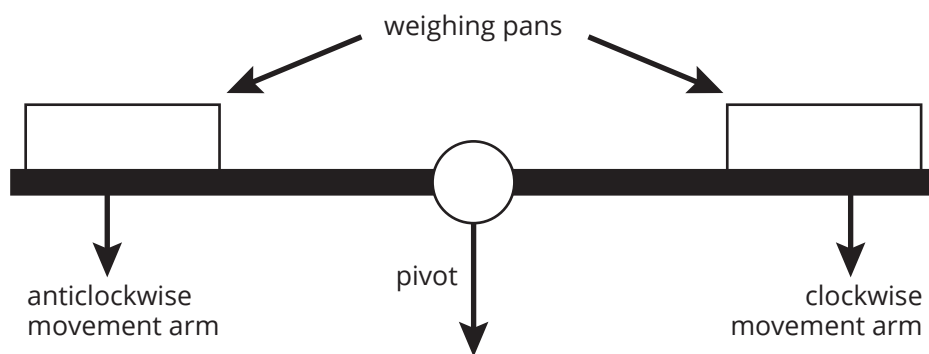
- b** Give Einstein's equation for the relationship between mass and energy. (1 mark)

Description	Marks
$E = mc^2$	1
Total	1

- c** The article states that 'mass ... might be considered to have a single quantum of energy'. Derive a mathematical expression to show how mass may be expressed to have a quantum of energy. (2 marks)

Description	Marks
Recognition of the need to use $E = hf$	1
$E = mc^2 = hf$ Thus, $m = \frac{hf}{c^2}$	1
Total	2

- d** Use a diagram to show the basis for the measurement of mass using the international prototype kilogram as described in the information given. Label the diagram to indicate the appropriate physics concepts for the basis of the measurement for a mass. Explain the principles on which this system operates to find the value of an unknown mass. (5 marks)



Description	Marks
Recognition that the principle of equal clockwise and anticlockwise moments about the pivot point is used to measure mass	1
Recognition that when arms of 'balance' are stationary and horizontal, the unknown mass will be equal to the known mass	1
Diagram shows equal length arms to left and right of pivot and pans same distance from pivot (or suitable representation for $\Sigma acm = \Sigma cm$)	1-2
Labels may include: • pivot • moment arms • pans	1
Total	5

- e** In paragraph six, what does the information state that mass may be expressed in terms of? (1 mark)

Description	Marks
Current measured against the force of gravity	1
Total	1

- f** Using a relationship in your study of electromagnetism, derive the mathematical expression for mass in terms of the variables given in paragraph six. (3 marks)

Description	Marks
$F = I\ell B$	1
$F = mg = I\ell B$	1
Thus, $m = \frac{I\ell B}{g}$	1
Total	3

- g** Using the information given in paragraph seven and the mathematical expression from (f), derive an expression for mass based on current and voltage. (3 marks)

Description	Marks
EMF, $V = \ell v B$	1
$B = \frac{V}{\ell v}$	1
Substituting for B in $m = \frac{I\ell B}{g}$ gives: $m = \frac{IV}{gv}$	1
Total	3